Apple vs Samsung

Competitive Financial & Market Analysis

Data Sources: Yahoo Finance (yfinance) | FRED (Federal Reserve)

Coverage: January 2015 – April 2026

Companies: Apple Inc. (AAPL, NASDAQ) | Samsung Electronics (005930.KS, KRX)

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Research Background & Questions

Research Background

Apple and Samsung are the world's two largest smartphone vendors and fierce competitors across hardware, software, and services. Analyzing their financial and market data provides insight into how competitive rivalry, macroeconomic forces, and business model differences shape investor outcomes over a full decade.

RQ 1

Stock Performance & Correlation

How have Apple and Samsung stock prices performed and correlated over 2015–2026? Do they move together or independently?

RQ 2

Financial Fundamentals

How do Apple and Samsung compare on key financial metrics — revenue growth, profitability margins, and free cash flow?

RQ 3

Macro Regression

Can macroeconomic factors (US CPI inflation and 10-year Treasury yield changes) explain monthly stock return variations?

Significance of the Study

Investor Decision-Making

Both stocks are held by millions of retail and institutional investors. Understanding their risk and return dynamics is directly actionable for portfolio construction.

Cross-Market Dynamics

Apple trades on NASDAQ (USD) and Samsung on KRX (KRW). Comparing them illuminates how U.S. and Korean markets respond differently to the same global events.

Business Model Contrast

Apple's high-margin services model versus Samsung's hardware-heavy semiconductor business makes this an ideal case study in strategic divergence.

Macro Sensitivity

The 2022–2023 rate-hike cycle had outsized impact on tech valuations. Studying which firm was more macro-exposed informs future investment strategy.

Data Collection

Source 1 — Yahoo Finance

via yfinance Python library | https://finance.yahoo.com

Tickers:	AAPL (NASDAQ) | 005930.KS (KRX)
Frequency:	Daily OHLCV stock prices
Coverage:	Jan 1, 2015 – Apr 2026
Financials:	Annual: income stmt, balance sheet, cash flow (4–5 years)
Key Vars:	Close, Revenue, Net Income, Gross Profit, Op. Income, Free Cash Flow

Source 2 — FRED (Federal Reserve)

via pandas-datareader | https://fred.stlouisfed.org

Series 1:	CPIAUCSL — CPI All Urban Consumers
Frequency:	Monthly (seasonally adjusted)
Series 2:	DGS10 — 10-Year Treasury Rate
Frequency:	Daily
Coverage:	Jan 2015 – Apr 2026
Key Vars:	CPI_US (inflation proxy), Treasury_10Y (rate level)

Data Cleaning & Manipulation Process

1 Index & Parsing

Converted date strings to DatetimeIndex; set date as index for all DataFrames and sorted chronologically.

2 Missing Value Handling

Identified NaN gaps (weekends/holidays, monthly CPI vs daily treasury). Dropped rows where both close prices were missing.

3 Data Type Conversion

Cast price/volume columns to float64. Converted financial statement values to numeric using pd.to_numeric(errors='coerce').

4 Subsetting & Filtering

Extracted specific rows from financial statements (Revenue, Net Income, Gross Profit, Operating Income). Filtered to 2015–2026.

5 New Column Generation

Computed daily returns (pct_change), profit margins, normalized prices (base=100), and annualized volatility (std × √252).

6 Merging & Resampling

Resampled daily prices to monthly (last day). Merged stock returns, CPI changes, and treasury yield changes via inner join on date.

7 Aggregation

Computed mean, median, min/max. Applied groupby on year for annual correlations. Calculated 90-day and 252-day rolling correlations.

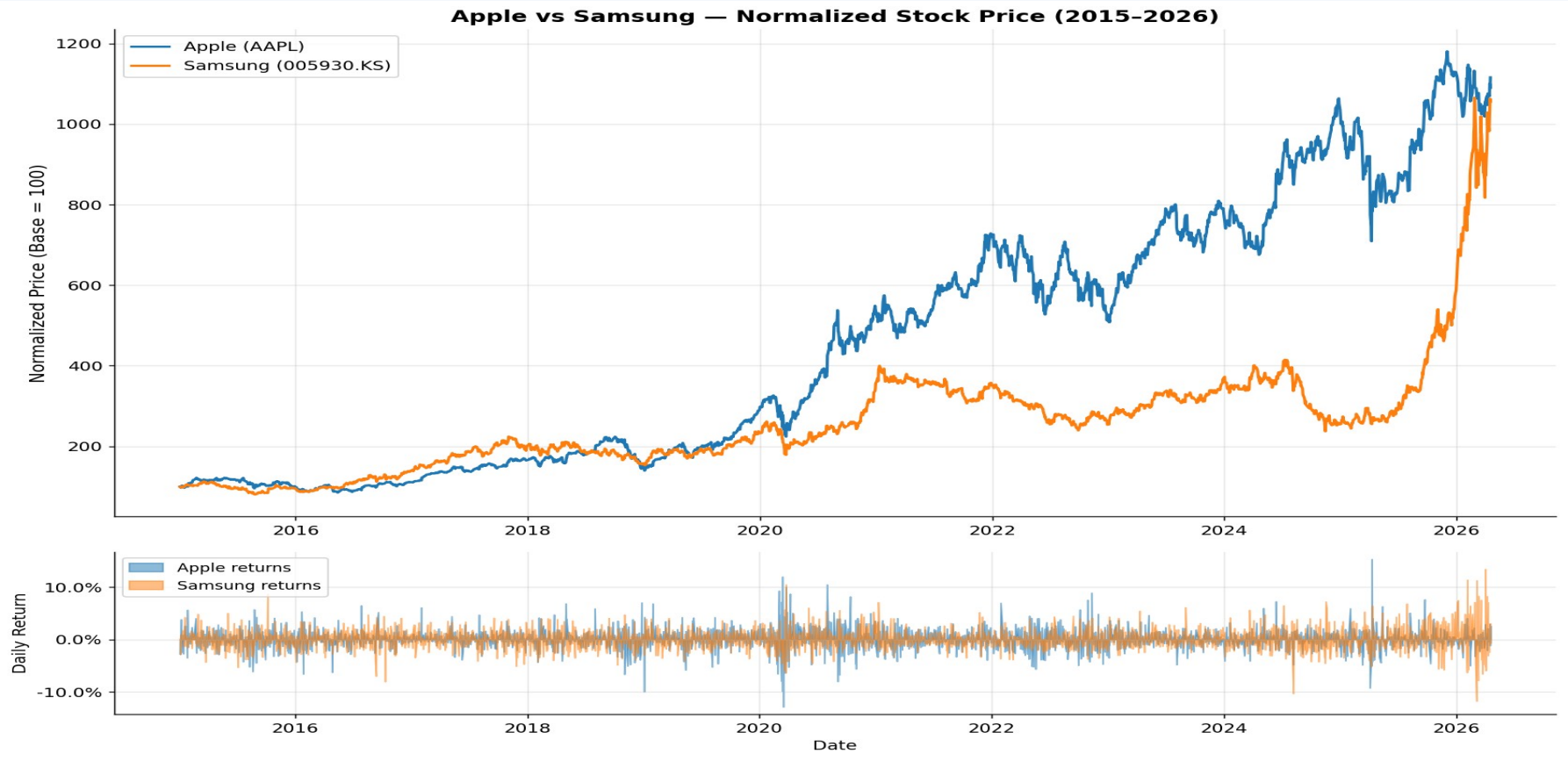
8 Scaling

Scaled Apple financials to USD billions (÷1e9), Samsung to KRW trillions (÷1e12) for readability in charts and tables.

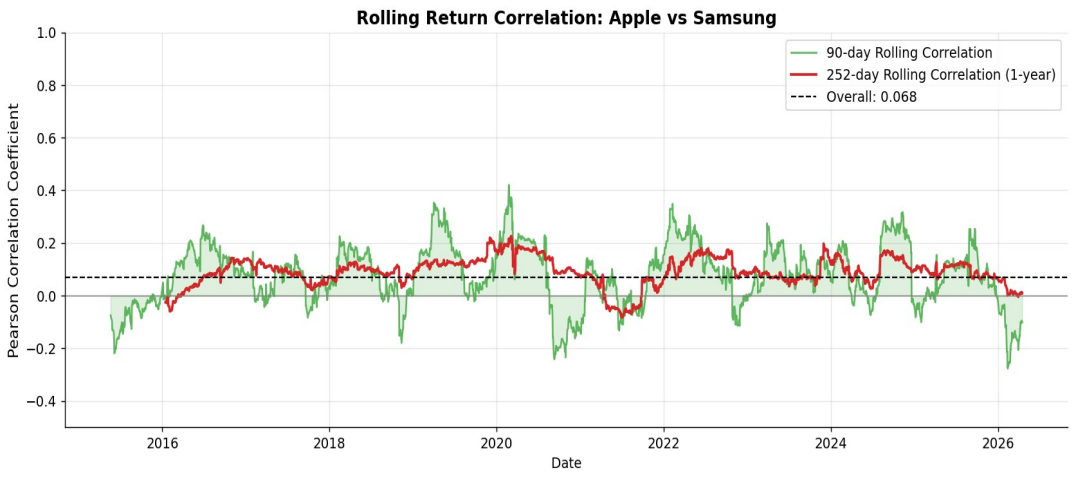
Analysis 1

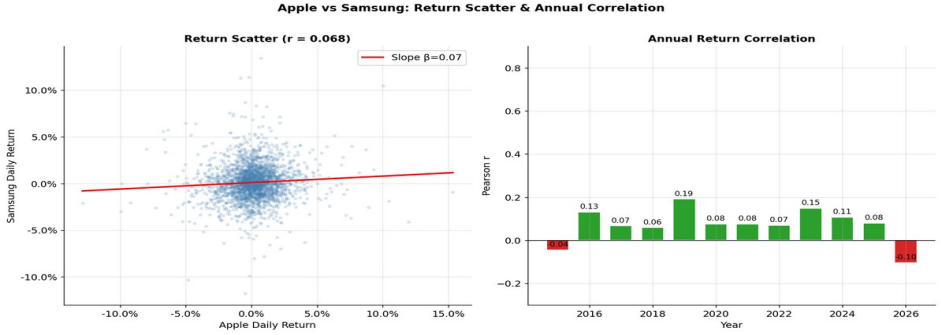
Stock Price Performance & Correlation

RQ 1 — Normalized Stock Price Performance (2015–2026)



RQ 1 — Return Correlation: Rolling & Annual Breakdown

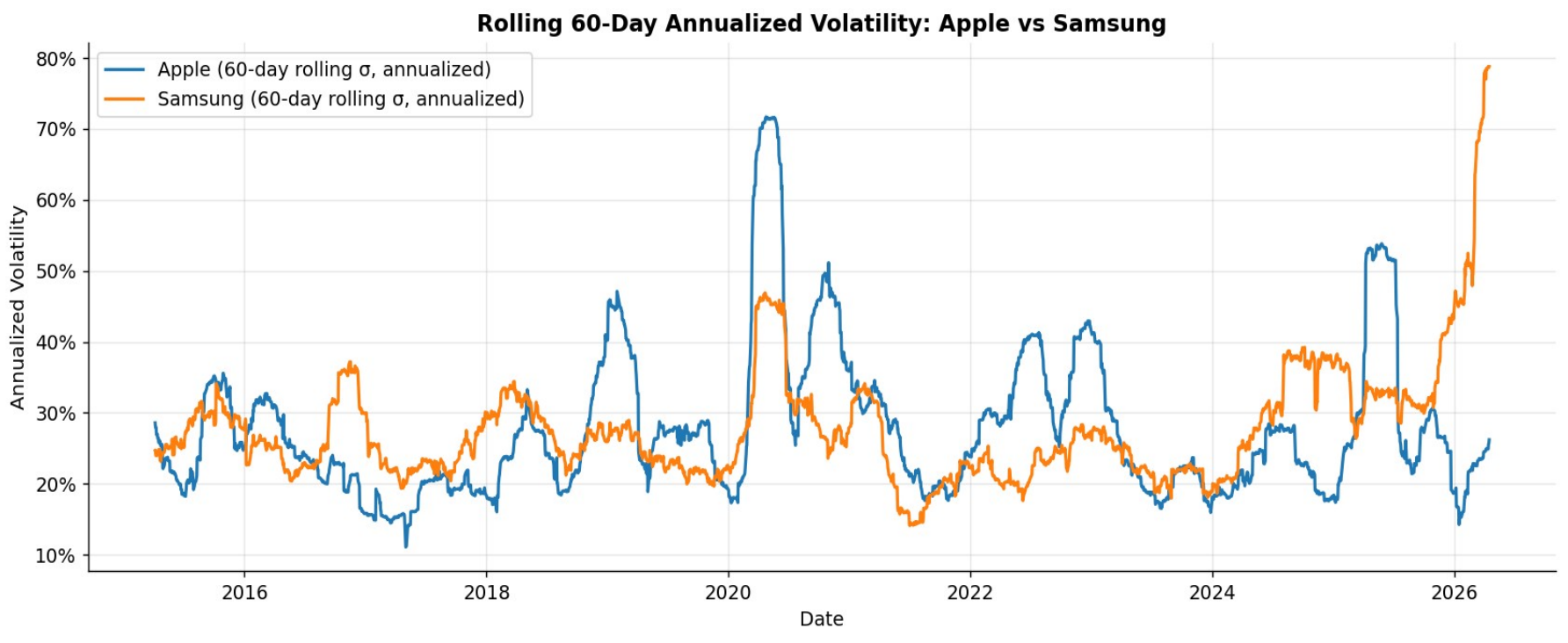




Key Findings

Overall r	**0.068 (very low)**
Peak corr.	**~0.45 (COVID 2020)**
Best year	**2020: r = 0.44**
Apple CAGR	**31.0% annualized**
Samsung CAGR	**30.5% annualized**
Apple vol.	**29.4% annualized**
Samsung vol.	**29.9% annualized**
Apple Sharpe	**1.05**
Samsung Sharpe	**1.02**

RQ 1 — Rolling Annualized Volatility

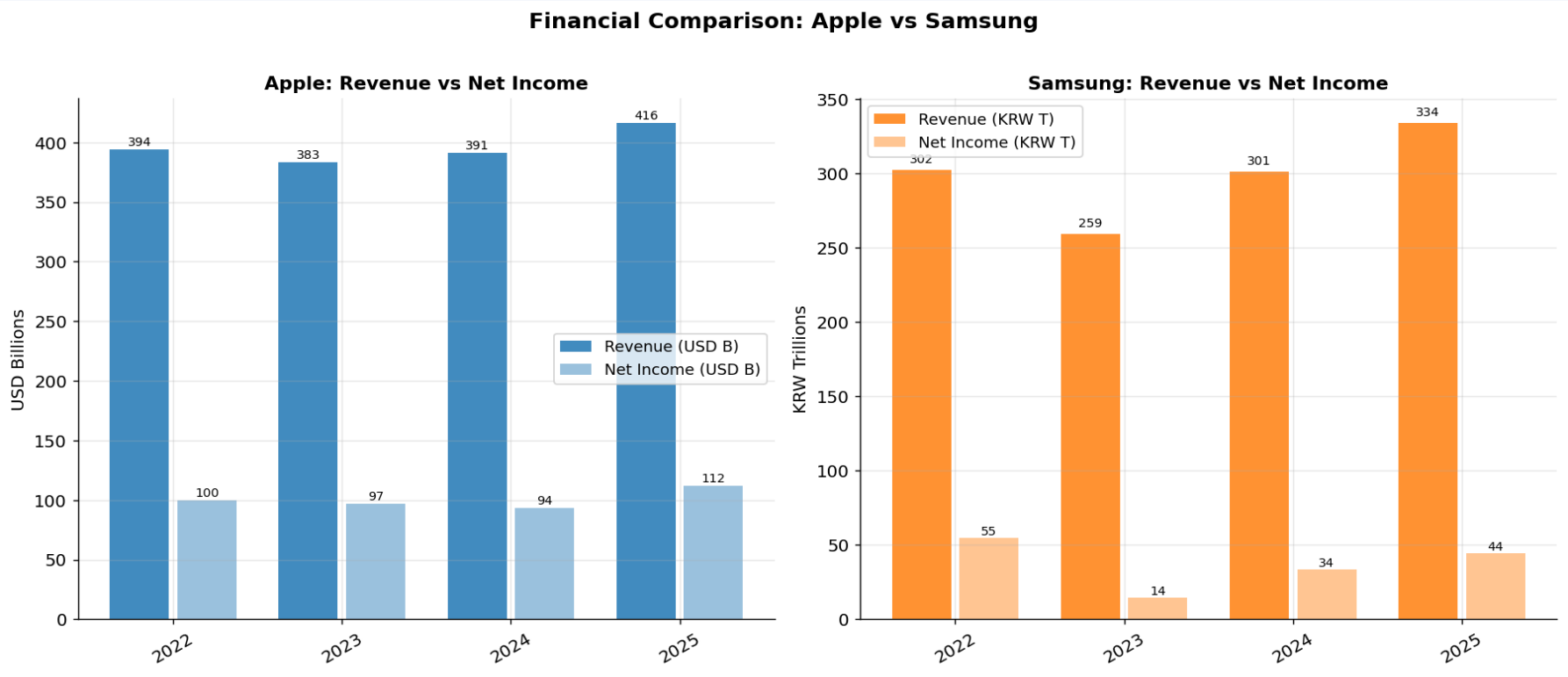


Both stocks maintain similar annualized volatility (~29–30%), with synchronized spikes during COVID-19 (March 2020) and the 2022 rate-hike cycle. Samsung shows slightly higher peak volatility during the 2023 semiconductor downturn.

Analysis 2

Financial Fundamentals Comparison

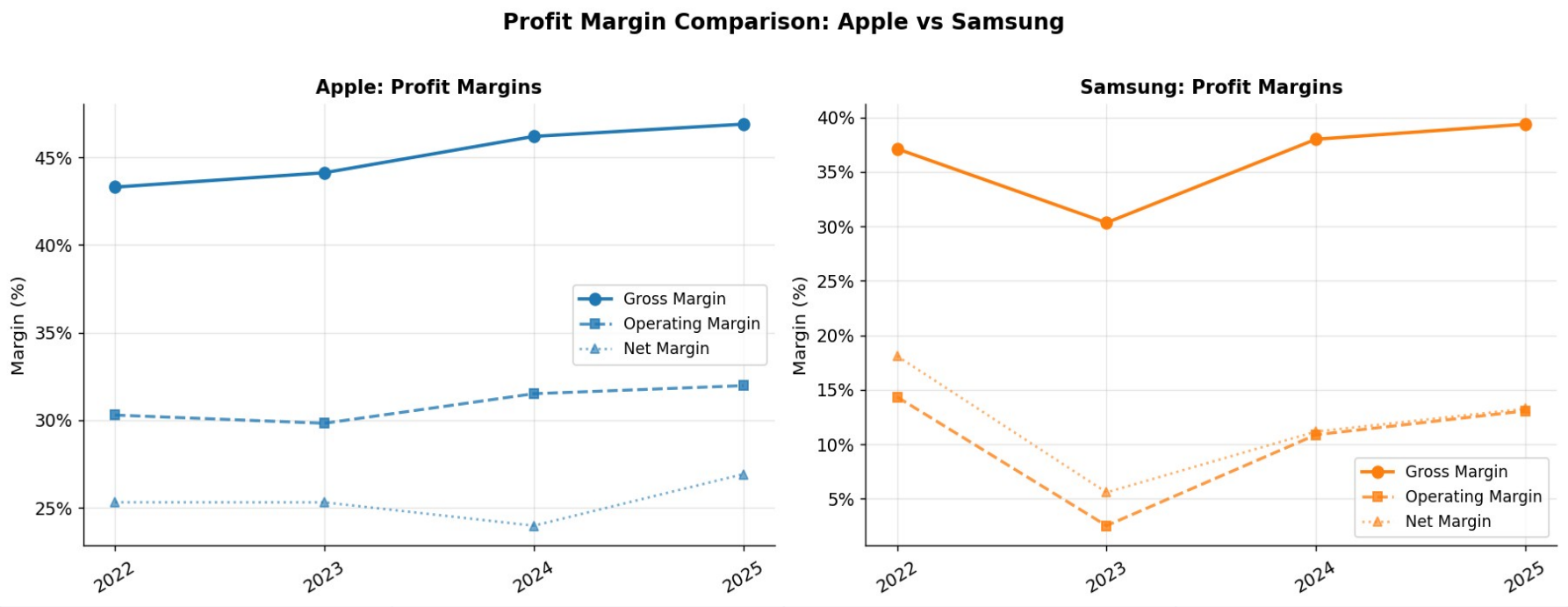
RQ 2 — Revenue & Net Income (2022–2025)



Apple: Revenue stable ~$383–416B USD. Strong net income ~$94–112B.

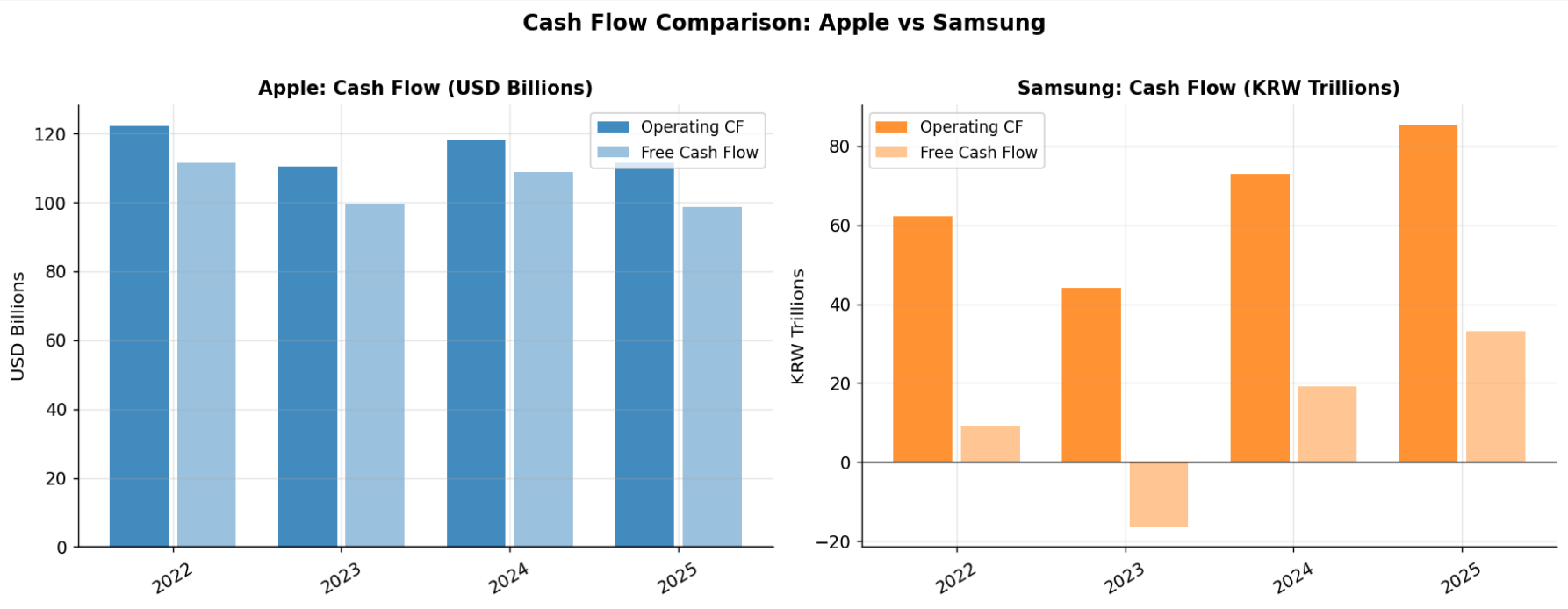
Samsung: Revenue dipped in 2023 (chip downturn) but recovered strongly to KRW 334T by 2025.

RQ 2 — Profit Margin Comparison



Metric	Apple Avg	Samsung Avg	Advantage
Gross Margin	**~47%**	~37%	Apple
Operating Margin	~31%	~8%	Apple
Net Margin	~25%	~12%	Apple

RQ 2 — Cash Flow Comparison



Apple FCF

~$100B/year consistently. Returns capital via buybacks ($90B+/yr). World-class cash generation relative to revenue.

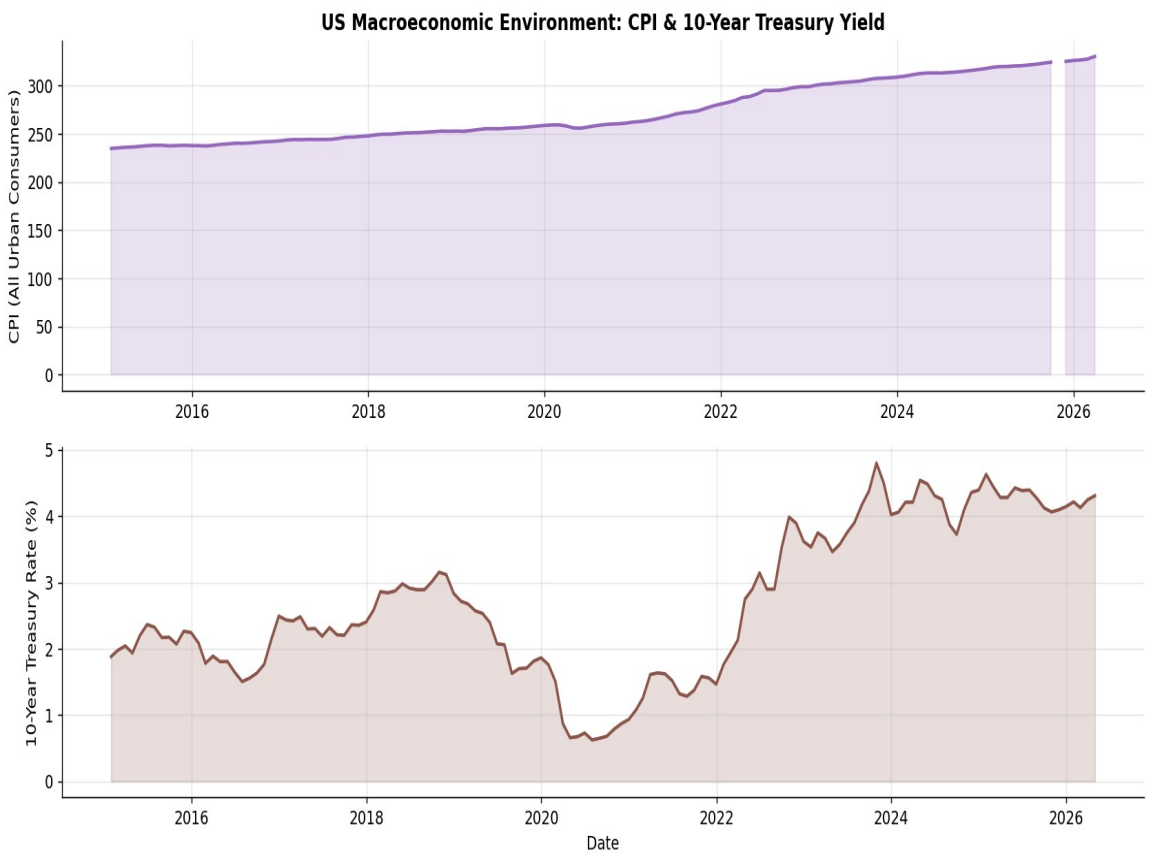
Samsung FCF

Highly volatile. Turned negative in 2023 due to massive chip fab investment. Recovered sharply to positive in 2025.

Analysis 3

Macroeconomic Regression Analysis

RQ 3 — US Macroeconomic Environment (2015–2026)



Macro Context

2015–2019

Low inflation (~1–2%), gradual Fed normalization. Favorable for tech.

2020

COVID: Fed cuts to 0%. CPI initially dipped then surged with stimulus.

2021–2022

Inflation peaked at 9%. Fed raised rates 525bps — tech repriced sharply.

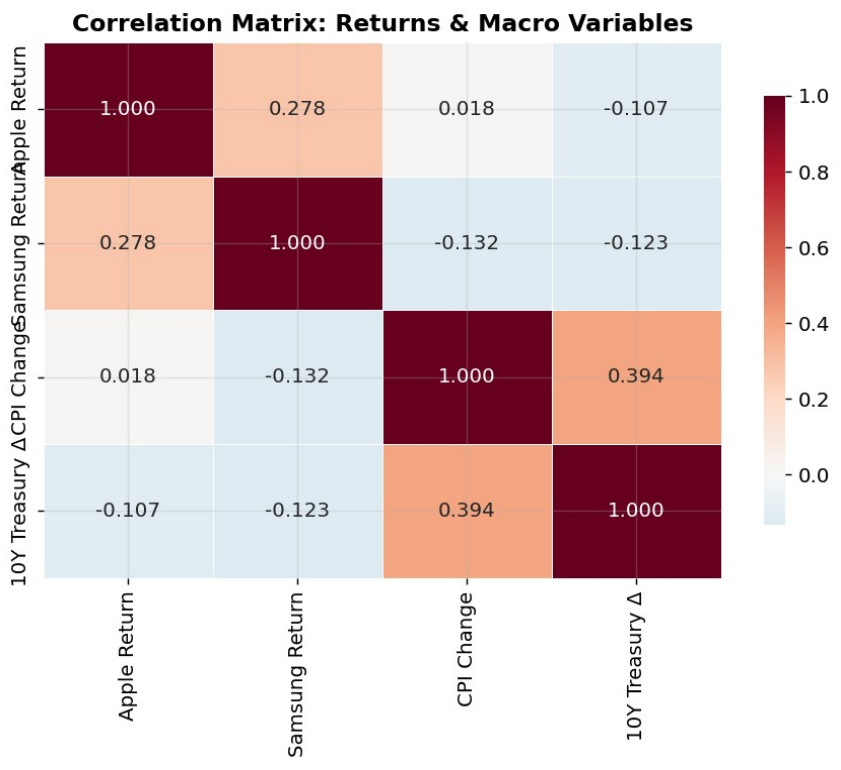
2023–2024

Inflation cooling. Fed pivot discussions. Tech stocks rebounded.

2025–2026

CPI near 3% target. Stable rates. Both stocks near all-time highs.

RQ 3 — Correlation Heatmap & Regression Setup

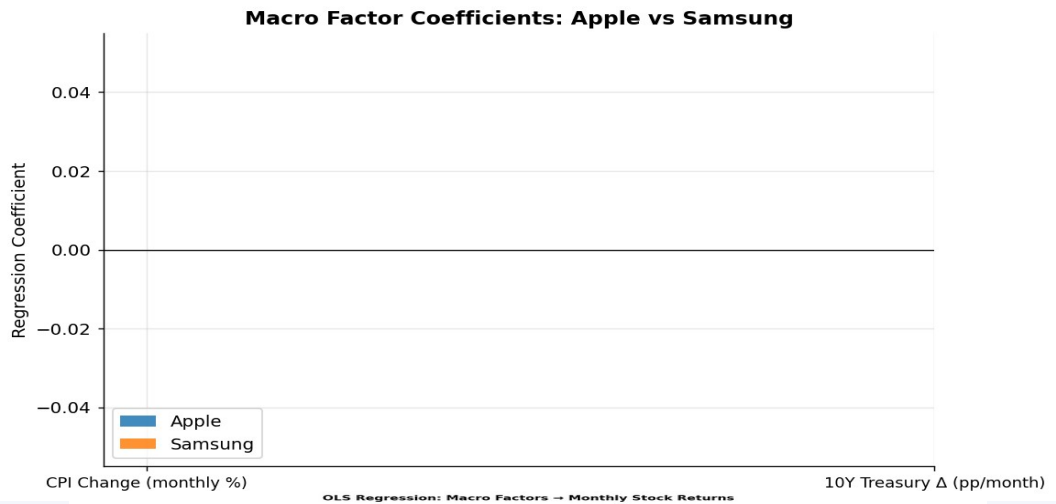


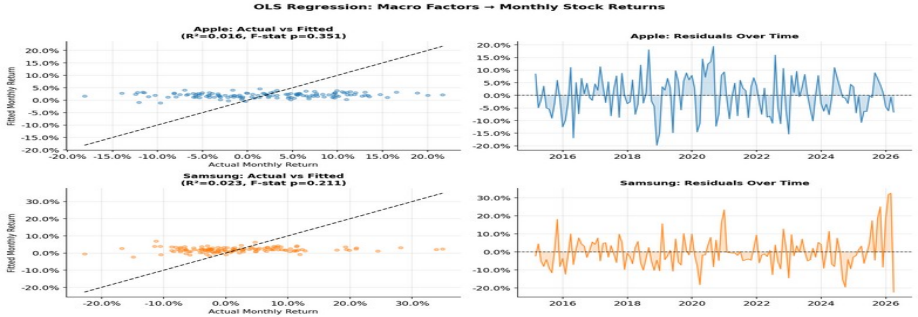
OLS Regression Setup

$$Monthly\ Return = \alpha + \beta_1 \times CPI_Chg + \beta_2 \times \Delta T10Y + \varepsilon$$

Dep. Var:	Monthly stock return (Apple or Samsung)
CPI_Chg:	Monthly % change in US CPI
ΔT10Y:	Monthly change in 10-yr Treasury (pp)
N:	134 monthly observations (2015–2025)
Method:	OLS via statsmodels (Python)
Hypothesis:	Rising inflation/rates reduce growth stock returns

RQ 3 — Regression Results





Regression Summary

Apple

R^2:	**0.016**
F-stat p:	**0.351**
CPI coef:	**+2.08 (p=0.45)**
10Y coef:	**−0.054 (p=0.15)**

Samsung

R^2:	**0.023**
F-stat p:	**0.211**
CPI coef:	**−3.36 (p=0.30)**
10Y coef:	**−0.040 (p=0.37)**

Low R^2 and non-significant F-stats confirm macro factors alone weakly predict returns. Firm-specific events dominate.

Discussion & Key Conclusions

RQ 1 — Stock Performance

- Both stocks: ~1,000% total return (2015–2026)
- Apple: 31.0% CAGR | Samsung: 30.5% — near-identical risk-adjusted returns
- Daily return correlation very low (r=0.068) despite being direct competitors
- Correlation rises during crises (COVID 2020) but time-varying
- Similar volatility (~29–30%), driven by different events

RQ 2 — Financial Fundamentals

- Apple outperforms on all margin metrics (gross ~47% vs Samsung ~37%)
- Apple's services segment boosts margin quality and reduces cyclicality
- Samsung driven by semiconductor cycles — sharp 2023 downturn, strong 2025 recovery
- Apple FCF ~$100B/yr; Samsung's is volatile and capital-intensive
- Both companies improved margins in FY2025

RQ 3 — Macro Regression

- Macro factors explain <3% of monthly return variance
- Coefficient signs align with theory (higher rates → lower Apple returns)
- Neither model statistically significant (F-stat p > 0.2)
- Firm-specific events (earnings, product launches) dominate
- A Fama-French model would substantially improve fit

Limitations & Future Work

Limitations

1. Currency Risk: Analysis doesn't adjust for USD/KRW exchange rate fluctuations, affecting cross-currency comparability.
2. Financial Statement Coverage: Only 4–5 years of annual financials available via yfinance, limiting long-term trend analysis.
3. Omitted Variables: Regression omits market beta, earnings surprises, and sector ETF flows.
4. Survivorship Bias: Both firms are industry leaders; findings may not generalize to broader tech sector.

Future Work

1. Apply Fama-French 3-factor model to control for market, size, and value risk premia.
2. Event study: analyze abnormal returns around major product launches (iPhone, Galaxy S-series).
3. Incorporate Google Trends data as a consumer interest leading indicator.
4. Extend to a panel of smartphone sector peers (Sony, Xiaomi, TSMC) for broader competitive analysis.

Team Member Roles & Contributions

Member	Role	Tasks
Member 1	Project Lead & Data Collection	Defined research questions; downloaded stock and financial data using yfinance; managed project repository
Member 2	Data Cleaning & EDA	Cleaned datasets; handled missing values; built descriptive statistics; created normalized price and volatility charts
Member 3	Analysis 1 — Correlation	Computed Pearson correlations; built rolling correlation and annual breakdown charts; wrote RQ1 conclusions
Member 4	Analysis 2 — Financials	Extracted financial statements; computed margins and FCF; designed financial comparison charts; wrote RQ2 conclusions
Member 5	Analysis 3 & Presentation	Built macro regression models; downloaded FRED data; created regression charts; assembled presentation slides

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Thank You

Questions & Discussion

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~1,000%

Total Return
Both Stocks

r = 0.068

Return
Correlation

47% vs 37%

Gross Margin
Apple vs Samsung